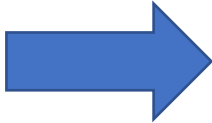


SAP RAP ABAP WORKSHOP WITH HANDS ON

DAY - 1

About Me?



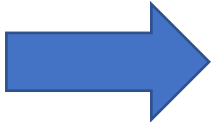
14+ Years of Real Time working Experience



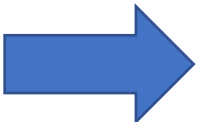
Worked in MNCs Like Accenture, SOPRASTERIA etc.



Implementation Projects - 2, Migration - 1, Rollout - 1 and Support Projects - 2



Helped 10000+ students to become better SAP technical consultant



Founder of SAPTECH Made Easy(www.saptechmadeeasy.com)



Owens an YouTube Channel which helps to

Agenda

RAP ABAP introduction

System Set up

One implementation scenario

- Creating Database table

- Create List report fiori application or preview using RAP ABAP

- Enable UI services

RAP ABAP

The ABAP RESTful Application Programming Model (RAP) allows developers to efficiently build cloud-ready, transactional business apps, services, and extensions on SAP BTP ABAP environment, SAP S/4HANA Cloud Public Edition, SAP S/4HANA Cloud Private Edition, and SAP S/4HANA 1909 and higher. RAP is at the heart of the ABAP Cloud development model for transactional development scenarios.

RAP: **R**estful **A**pplication **P**rogramming

ABAP ON CLOUD

Cloud: BTP

BTP: Business Technology Platform

(HCP→SCP→BTP)

ECC/S4HANA(On premise)

ABAP

@ODATA.PUBLISH: TRUE

ZCDSDEMO_CDS

CRUD

BOPF:

RAP :

RAP has two ways of developing the application

1. Managed Scenario (CRUD)
2. Unmanaged Scenario (Custom logic, custom code)

Table:

CDS: (Root view, and Projection view)

Behaviour Definition

Behavior Implementation

Service definition

Service binding

ODATA SERVICE(ODATA V2 , ODATA V4)

WEB API

Global Account (TATA GLOBAL ACCOUNT)

--SRM_SUBACCOUNT 1

--MDG_SUBACCOUNT 2

--FI/CO_SUBACCOUNT 3

SAP RAP ABAP WORKSHOP WITH HANDS ON

DAY - 2

Enabling the field for filter

Travel ID:

Agency ID:

Customer ID:

Annotations

```
@UI.selectionField: [ { position: <position number> } ]
```

Example:

```
@UI.selectionField: [ { position: 10 } ]
```

@UI.lineItem: [{ position: <postion number> }]

Example:

@UI.lineItem: [{ position: 10 }]

→ ↻ 🏠 <https://11b69711-9bcb-491b-b504-a9ba7e4b7e97.abap-web.us10.hana.ondemand.com/sap/bc/adt/businessservices/odatav4/feap/C...C...C...C...xuHC...C...%7Ds...> ☆ ⚙️ 👤

SAP Preview for Fiori Elements App ▾

Standard * ▾

Editing Status: ▾ Travel ID: 📄 Agency ID: 📄 Customer ID: 📄 Starting Date: 📄

Search 🔍 All ▾

These fields comes automatically as column header with below annotation
@UI.lineItem: [{ position: 40 }]

Go Adapt Filters (1) Enter

Create Delete 📄 ⌵ ⚙️ 📄 | ▾

☐	Agency ID	Agency Name	Booking Fee	Changed By	Changed On	Currency Co...	Customer ID	
☐	70041	Maxitrip	40.00	USD	CB9980005574	May 28, 2025, 10:58:10 PM	USD	594 >
Description: Vacation								
☐	70007	srrrr	20.00	USD			USD	608 >
Description: Business Trip for Christine, Pierre								
☐	70046	Hendrik's	80.00	USD			USD	93 >
Description: Vacation								
☐	70042	The Ultimate Answer	40.00	USD	CB9980005574	May 28, 2025, 11:00:08 PM	USD	665 >
Description: Vacation								
☐	70007	srrrr	20.00	USD			USD	225 >

To display object page:

```
@UI.facet: [ {  
  label: 'General Information',  
  id: 'GeneralInfo',  
  purpose: #STANDARD,  
  position: 10 ,  
  type: #IDENTIFICATION_REFERENCE  
} ]
```

Object Page



###GENERATED Core Data Service Entity ▾



1

1

Edit

Delete



Ctrl+E

General Information

Travel ID:

1

Agency ID:

70041

To display the field on the object page

```
@UI.identification: [ { position: <position number> } ]
```

 ###GENERATED Core Data Service Entity 

1

1

Edit

Delete

 ▼

Ctrl+E

General Information

Travel ID:
1

Agency ID:
70041

```

1 @AccessControl.authorizationCheck: #CHECK
2 @Metadata.allowExtensions: true
3 @EndUserText.label: '###GENERATED Core Data Service Entity'
4 @ObjectModel.sapObjectNodeType.name: 'ZRAP100_PANKAJ85'
5 @ObjectModel.semanticKey: [ 'TravelId' ]
6 define root view entity ZR_RAP100_PANKAJ85
7   as select from zrap100 pankaj85 as travel
8   association [0..1] to /DMO/I_Agency          as _Agency      on $projection.AgencyID = _Agency.AgencyID
9   association [0..1] to /DMO/I_Customer        as _Customer    on $projection.CustomerID = _Customer.CustomerID
10  association [1..1] to /DMO/I_Overall_Status_VH as _OverallStatus on $projection.OverallStatus = _OverallStatus.OverallStatus
11  association [0..1] to I_Currency              as _Currency    on $projection.CurrencyCode = _Currency.Currency
12
13 {
14   key travel_id          as TravelId,
15   agency_id              as AgencyID,
16   customer_id            as CustomerID,
17   begin_date              as BeginDate,
18   end_date                as EndDate,
19   @Semantics.amount.currencyCode: 'CurrencyCode'
20   booking_fee             as BookingFee,
21   @Semantics.amount.currencyCode: 'CurrencyCode'
22   total_price             as TotalPrice,
23   @Consumption.valueHelpDefinition: [ {
24     entity.name: 'I_CurrencyStdVH',
25     entity.element: 'Currency',

```


Use like below

```
key TravelID,  
  AgencyID,  
  _Agency.Name as AgencyName,  
  CustomerID,  
  _Customer.LastName as CustomerName,  
  BeginDate,  
  EndDate,  
  BookingFee,  
  TotalPrice,  
  CurrencyCode,  
  Description,  
  OverallStatus,  
  _OverallStatus._Text.Text as OverallStatusText : localized,  
  Attachment,  
  MimeType,  
  FileName,  
  LastChangedAt
```

Exposed Association

```
8 @Semantics.user.localInstanceLastChangedBy: true
9 local_last_changed_by as LocalLastChangedBy,
0 @Semantics.systemDateTime.localInstanceLastChangedAt: true
1 local_last_changed_at as LocalLastChangedAt,
2 @Semantics.systemDateTime.lastChangedAt: true
3 last_changed_at as LastChangedAt,
4 _Agency,
5 _Currency,
6 _OverallStatus,
7 _Customer
```

Enable `@Search.searchable: true` at header level

```
1 @Metadata.allowExtensions: true
2 @EndUserText.label: '###GENERATED Core Data Service Entity'
3 @AccessControl.authorizationCheck: #CHECK
4 @ObjectModel.sapObjectType.name: 'ZRAP100_PANKAJ85'
5 @Search.searchable: true
6 @ObjectModel.semanticKey: [ 'TravelId' ] //makes bold
7 define root view entity ZC_RAP100_PANKAJ85
8   provider contract transactional_query
9   as projection on ZR_RAP100_PANKAJ85 as travel
10 {
11
```



It enables to search

When this search is enabled at least one element should have default search enabled to be able to work this search

Annotation to enable the search at field label is:

`@Search.defaultSearchElement: true`

Enabling annotation at element of text type will make more sense,
However you can enable for other data type as well

Enable search help on the fields

At least one such annotation is required to enable the search

```
[TRL] ZRAP1... [TRL] ZC_RA... x [TRL] ZRAP1... [TRL] ZRAP1... [TRL] ZRAP1... [TRL] ZRAP1... [TRL] ZR_RA... [T]
7 define root view entity ZC_RAP100_PANKAJ85
8   provider contract transactional_query
9   as projection on ZR_RAP100_PANKAJ85 as travel
10 {
11
12   @Search.defaultSearchElement: true
13   // @Search.fuzzinessThreshold: 0.90
14   key TravelId,
15
16   @Search.defaultSearchElement: true
17   // @Search.fuzzinessThreshold: 0.90
18   // @ObjectModel.text.element: ['AgencyName']
19   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_Agency_StdVH', element: 'Agenc
20   AgencyID,
21
22   @Search.defaultSearchElement: true
23   // @Search.fuzzinessThreshold: 0.90
24   _Agency.Name as AgencyName,|
25
26   @Search.defaultSearchElement: true
27   // @ObjectModel.text.element: ['lastname']
28   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_customer_StdVH', element: 'Cus
29   CustomerID,
30   // @Search.fuzzinessThreshold: 0.90
31   @Search.defaultSearchElement: true
32   _Customer.FirstName as firstName,
33   // @Search.fuzzinessThreshold: 0.90
34   @Search.defaultSearchElement: true
35   _Customer.LastName as lastname,
36
37   @Search.defaultSearchElement: true
38   BeginDate,
39   EndDate
```

Enable Search tolerance

@Search.fuzzinessThreshold:<matching percentage>

```
7 define root view entity ZC_RAP100_PANKAJ85
8   provider contract transactional_query
9   as projection on ZR_RAP100_PANKAJ85 as travel
10 {
11
12   @Search.defaultSearchElement: true
13   // @Search.fuzzinessThreshold: 0.90
14   key TravelId,
15
16   @Search.defaultSearchElement: true
17   @Search.fuzzinessThreshold: 0.90
18   // @ObjectModel.text.element: ['AgencyName']
19   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_Agency_StdVH', element: 'Agency
20   AgencyID,
21
22   @Search.defaultSearchElement: true
23   @Search.fuzzinessThreshold: 0.90
24   _Agency.Name as AgencyName,
25
26   @Search.defaultSearchElement: true
27   // @ObjectModel.text.element: ['lastname']
28   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_customer_StdVH', element: 'Cust
29   CustomerID,
30   @Search.fuzzinessThreshold: 0.90
31   @Search.defaultSearchElement: true
32   _Customer.FirstName as firstName,
33   @Search.fuzzinessThreshold: 0.90
34   @Search.defaultSearchElement: true
35   _Customer.LastName as lastname,
36
37   @Search.defaultSearchElement: true
38   BeginDate,
39   EndDate
```

Bold the primary key for user to mark that it's a key field

```
@ObjectModel.semanticKey: [ '<key field>' ]
```

```
1 @Metadata.allowExtensions: true
2 @EndUserText.label: '###GENERATED Core Data Service Entity'
3 @AccessControl.authorizationCheck: #CHECK
4 @ObjectModel.sapObjectType.name: 'ZRAP100_PANKAJ85'
5 @Search.searchable: true
6 @ObjectModel.semanticKey: [ 'TravelId' ] //makes bold
7 define root view entity ZC_RAP100_PANKAJ85
8   provider contract transactional_query
9   as projection on ZR_RAP100_PANKAJ85 as travel
10 {
11
12   @Search.defaultSearchElement: true
13   // @Search.fuzzinessThreshold: 0.90
14   key TravelId,
```

Bold the primary key for user to mark that it's a key field

@ObjectModel.semanticKey: ['TravelId']



☐ Travel ID	Agency ID	Customer ID	Starting Date	
☐ 1	Maxitrip (70041)	Illya (594)	Mar 16, 2025	>
☐ 2	srrrr (70007)	Benjamin (608)	Mar 8, 2025	>
☐ 3	Hendrik's (70046)	Sophie (93)	Mar 8, 2025	>
☐ 4	The Ultimate Answer (70042)	Harish (665)	Mar 8, 2025	>
☐ 5	srrrr (70007)	August (225)	Mar 8, 2025	>
☐ 6	Flights and More (70049)	Theresia (72)	Mar 8, 2025	>
☐ 7	Hendrik's (70046)	August (138)	Mar 8, 2025	>
☐ 8	Ben McCloskey Ltd. (70012)	Lee (705)	Mar 8, 2025	>
☐ 9	Sunny, Sunny, Sunny (70032)	Laura (115)	Mar 8, 2025	>

Enable human readable Text field in bracket for number type field

@ObjectModel.text.element:['<text field name>']

```
7 define root view entity ZC_RAP100_PANKAJ85
8   provider contract transactional_query
9   as projection on ZR_RAP100_PANKAJ85 as travel
10 {
11
12   @Search.defaultSearchElement: true
13   // @Search.fuzzinessThreshold: 0.90
14   key TravelId,
15
16   @Search.defaultSearchElement: true
17   @Search.fuzzinessThreshold: 0.90
18   @ObjectModel.text.element:['AgencyName']
19   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_Agency_StdVH', element: 'Agency
20   AgencyID,
21
22   @Search.defaultSearchElement: true
23   @Search.fuzzinessThreshold: 0.90
24   _Agency.Name as AgencyName,
25
26   @Search.defaultSearchElement: true
27   @ObjectModel.text.element:['firstName']
28   // @Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_customer_StdVH', element: 'Cust
29   CustomerID,
30   @Search.fuzzinessThreshold: 0.90
31   @Search.defaultSearchElement: true
32   _Customer.FirstName as firstName,
33   @Search.fuzzinessThreshold: 0.90
34   @Search.defaultSearchElement: true
35   _Customer.LastName as lastname,
36
37   @Search.defaultSearchElement: true
38   BeginDate,
39   EndDate
```


Enable human readable Text field in bracket for number type field
`@ObjectModel.text.element:['<text field name>']`



Preview for Fiori Elements App ▾

Standard ▾

Search Editing Status: Travel ID: Agency ID: Customer ID: Startii



Create Del

☐	Travel ID	Agency ID	Customer ID	Starting Date	
☐	1	Maxitrip (70041)	Illya (594)	Mar 16, 2025	>
☐	2	srrrr (70007)	Benjamin (608)	Mar 8, 2025	>
☐	3	Hendrik's (70046)	Sophie (93)	Mar 8, 2025	>
☐	4	The Ultimate Answer (70042)	Harish (665)	Mar 8, 2025	>
☐	5	srrrr (70007)	August (225)	Mar 8, 2025	>
☐	6	Flights and More (70049)	Theresia (72)	Mar 8, 2025	>
☐	7	Hendrik's (70046)	August (138)	Mar 8, 2025	>
☐	8	Ben McCloskey Ltd. (70012)	Lee (705)	Mar 8, 2025	>
☐	9	Sunny, Sunny, Sunny (70032)	Laura (115)	Mar 8, 2025	>



Front End Validations

Front end validations are defined like below –

```
@Consumption.valueHelpDefinition.useForValidation: true
```

Enable search help

```
@Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_customer_StdVH',  
element: 'CustomerID' } }
```

Before Search help annotation

Define Conditions: Customer ID

Equal to

Value



Add

No Conditions Entered



OK

Cancel



After search help annotations

Customer ID

Search and Select

Define Conditions

Search

Go

Show Filters

Items (728)

☐	Customer ID	First Name	Last Name	Title	
☐	1	Theresia	Buchholm	Mrs.	L
☐	2	Johannes	Buchholm	Mr.	M
☐	3	James	Buchholm	Mr.	F
☐	4	Thomas	Buchholm	Mr.	H
☐	5	Anna	Buchholm	Mrs.	H
☐	6	Uli	Buchholm	Mrs.	C
☐	7	Jean-Luc	Buchholm	Mr.	L
☐	8	August	Buchholm	Mr.	L

No Items or Conditions Selected

OK

Cancel

Customer ID:

Starting Date:

Go

Adapt Filters (1)

Create

Delete

Date

SAP
Tech
Made
Easy

To start, set the relevant filters and choose "Go".

Front end validation

```
@Consumption.valueHelpDefinition: [{ entity: {name: '/DMO/I_Agency_StdVH',  
element: 'AgencyID' } , useForValidation: true } ]
```

While creating or changing field on the object page, value will be validated

SAP RAP ABAP WORKSHOP WITH HANDS ON

DAY - 3

Enabling Attachment

```
currency_code      as CurrencyCode,  
description        as Description,  
overall status    as OverallStatus,  
@Semantics.largeObject: { mimeType: 'MimeType',    //case-sensitive  
                           fileName: 'FileName',    //case-sensitive  
                           acceptableMimeTypes: ['image/png', 'image/jpeg'],  
                           contentDispositionPreference: #ATTACHMENT }  
attachment         as Attachment,  
@Semantics.mimeType: true  
mime_type          as MimeType,  
file_name          as FileName,
```

Value for Mime Type and File Name is Case sensitive

`contentDispositionPreference`: How content should be displayed

SAP RAP ABAP

WORKSHOP

WITH HANDS ON

DAY - 4

From Timing perspective

Numbering: **Setting value for primary key at runtime**

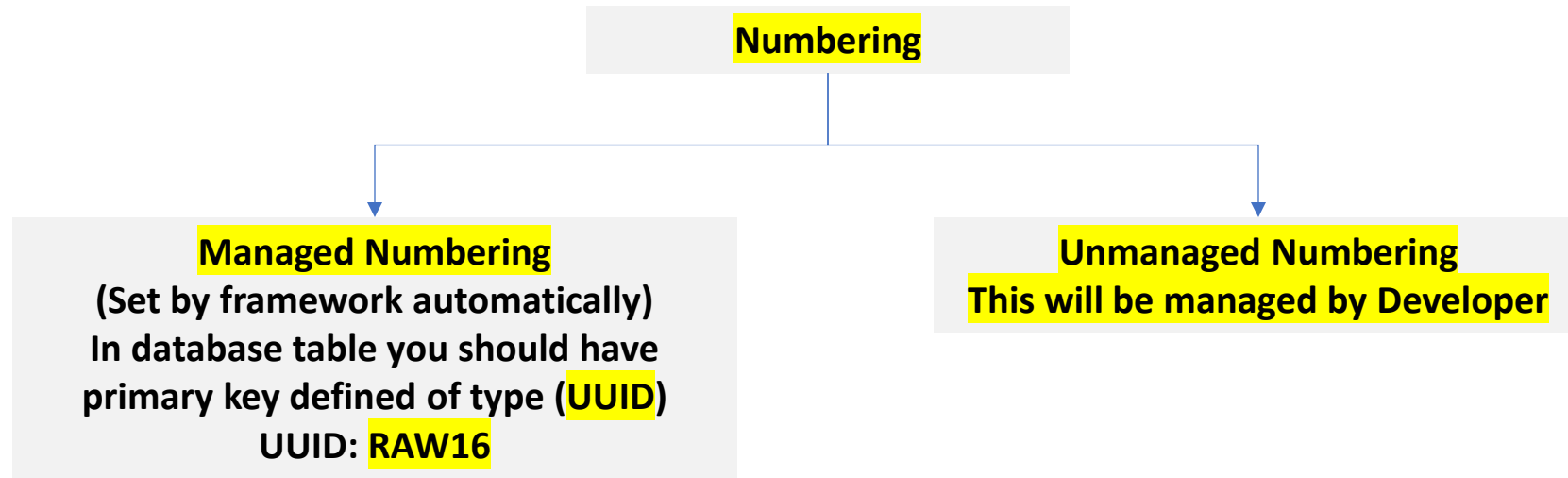
Early Numbering:

- Set by consumer(**External Numbering**)
- or by framework just after the modify or create request
- Or by implementing the **FOR NUMBERING** method

Late Numbering:

- Not set by consumer
- Set by internal framework after the point of no return of interaction phase is passed and SAVE sequence is triggered

From creation perspective



```
1 managed implementation in class ZRAP100_BP_TravelTP_000 unique;
2 strict ( 2 );
3 with draft;
4
5 define behavior for ZRAP100_R_TRAVELTP_000 alias Travel
6 persistent table zrap100_atrav000
7 draft table ZRAP100_DTRAV000
8 etag master LastChangedAt
9 lock master total etag LocalLastChangedAt
10 authorization master( global )
11 early numbering
12 {
13     field ( readonly )
14         CreatedAt,
15         CreatedBy,
16         LocalLastChangedAt,
17         LastChangedAt,
18         LastChangedBy;
19
20
21 create;
22 Early Numbering for CREATE ZRAP100_R_TRAVELTP_000 is not implemented
23
24
```

How to add extra field as input in popup while creating travel record

Automatic Early Numbering Implementation through Code

There are two way in which we can implement the automatic early numbering:

We can use the number range object(SNRO)

/DMO/TRV_M